

Homework 2

1. The 2D heat problem on $(0, \pi)^2$ with homogeneous Dirichlet boundary conditions has the forward operator

$$Kx(t_1, t_2) = \int_0^\pi \int_0^\pi k(t_1, t_2, s_1, s_2)x(s_1, s_2)ds_1ds_2$$

with kernel

$$k(t_1, t_2, s_1, s_2) = \sum_{n,m} e^{-(n^2+m^2)T} \sin(ns_1) \sin(ms_2) \sin(nt_1) \sin(mt_2).$$

It takes as input the image $x(s_1, s_2)$ as initial condition and outputs the heat distribution at time T , $y(t_1, t_2) = Kx(t_1, t_2)$.

- (a) Describe in an essay text (max 10 lines) why the 2D backwards heat equation is ill-posed.
 - (b) Implement numerically using FFT the operator K as a Matlab/Python/Julia... function that takes as input an N by N matrix x of sampled values of the function x and returns an N by N matrix y approximation $y(t_1, t_2)$. Show that the function acts correct on $x(s_1, s_2) = \sin(ks_1) \sin(ls_2)$.
 - (c) Download the image *KarinHarbo.jpg* and create from pixels in the image a 512 by 512 matrix x . Fix $T = 0.001$ and compute $y = Kx$. Add random Gaussian noise to y of magnitude $\delta = 0.01$ to find y^δ . Display y^δ .
 - (d) Show that K is selfadjoint. Solve the inverse problem using the Landweber iteration (take 500 iterations).
2. Consider the Abel problem for $x(s) = 1/2, y(t) = \sqrt{t}$. Discretize K by a matrix $A \in \mathbb{R}^{256 \times 256}$ and discretize similarly $x(s), y(t)$ by $x, y \in \mathbb{R}^{256}$. Add random noise of magnitude 10% to y to get y^δ . Denote by L_1 a finite difference matrix approximating the derivative.
 - (a) Show that $x \in H^1(0, \pi)$. Thus it makes sense to apply H^1 regularization.
 - (b) We approximate the H^1 -norm by $\|x\|_2^2 + \|L_1x\|_2^2$. Show that the problem

$$x_\alpha = \arg \min_x \|Ax - y^\delta\|_2^2 + \alpha(\|x\|_2^2 + \|L_1x\|_2^2) \quad (1)$$

has a unique solution for any $\alpha > 0$.

- (c) Find the solution to the ordinary Tikhonov problem (without L_1) with α obtained by the discrepancy principle.
- (d) Find the solution to the problem (1), again choosing α by the discrepancy principle. Compare this solution to the previous without L_1 .

Important notes:

- Deadline for the HW2 is November 12.
- All homeworks must be approved in order to qualify for the exam.
- Your background is different, and therefore you can choose to put emphasis differently in your solution.
- Demonstrate in your solution your understanding of the material and computational competencies.
- Each student should hand in a single pdf via DTU Learn. The number of pages for the main report must not exceed 6. Code can be put in an appendix.
- Comment on all results: What do you see, and why is it reasonable (or unreasonable)?
- Make quality plots with proper names on axis, legends and captions.